

§7.1 - Integration by Parts

Definition #1 (Real numbers)

The set of all **real numbers** is denoted by

$\mathbf{R}$.

If x is a real number we express this as

$x \in \mathbf{R}.$

Definition #2

We also have the sets of **natural numbers**, **integers**, and **rational numbers**:

$$\mathbf{N} = \{1, 2, 3, \dots\}$$

$$\mathbf{Z} = \{0, \pm 1, \pm 2, \pm 3, \dots\}$$

$$\mathbf{Q} = \{m/n \mid m \in \mathbf{Z} \text{ and } n \in \mathbf{N}\}.$$

Observe

$$\mathbf{N} \subset \mathbf{Z} \subset \mathbf{Q} \subset \mathbf{R}.$$

Definition #3 (Antiderivative)

Let F and f be functions on an interval I .

If

$$F'(x) = f(x)$$

on I then F is an **antiderivative** of f . We denote this as

$$\int f(x) \, dx = F(x)$$

Theorem #1

Let F be any antiderivative of the function f on an interval I .

Then all of f 's antiderivatives are of the form

$$F(x) + C$$

where $C \in \mathbf{R}$.

Theorem #2 (Fundamental Theorem of Calculus)

Let f be continuous on the interval $[a, b]$ and define g by

$$g(x) = \int_a^x f(t) \, dt.$$

Then

1. g is continuous on $[a, b]$ and g' exists on (a, b) where

$$g'(x) = f(x)$$

2. g is an antiderivative of f on $[a, b]$.

3. We have

$$F(b) - F(a) = \int_a^b f(x) \, dx$$

where F is any antiderivative of f on $[a, b]$.

Remark

Some shorthand:

$$F(b) - F(a) = F(x)]_a^b = [F(x)]_a^b = F(x)|_{x=a}^b$$

Theorem #3 (Chain Rule)

Let g be differentiable at x and f be differentiable at $g(x)$.

Then

$$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x).$$

Theorem #4 (Integration by Substitution)

Let $u = g(x)$ be a differentiable function whose range is an interval I .

If f is continuous on I then

$$\int f(g(x)) \cdot g'(x) \, dx = \int f(u) \, du.$$

Example #1

Find

$$\int \frac{x}{x-3} dx.$$

Theorem #5 (Integration by Substitution for Definite Integrals)

Let g' be continuous on the interval $[a, b]$.

If f is continuous on the range of $u = g(x)$ then

$$\int_a^b f(g(x)) \cdot g'(x) \, dx = \int_{g(a)}^{g(b)} f(u) \, du.$$

Theorem #6 (Product Rule)

Let u and v be differentiable functions.

Then

$$\frac{d}{dx} (u(x) \cdot v(x)) = u'(x) \cdot v(x) + u(x) \cdot v'(x).$$

Theorem #7 (Integration by Parts)

Let u and v be differentiable functions.

Then

$$\int u \, dv = u \cdot v - \int v \, du$$

Example #2

Find

$$\int 3xe^x \, dx$$

Theorem #8 (Integration by Parts for Definite Integrals)

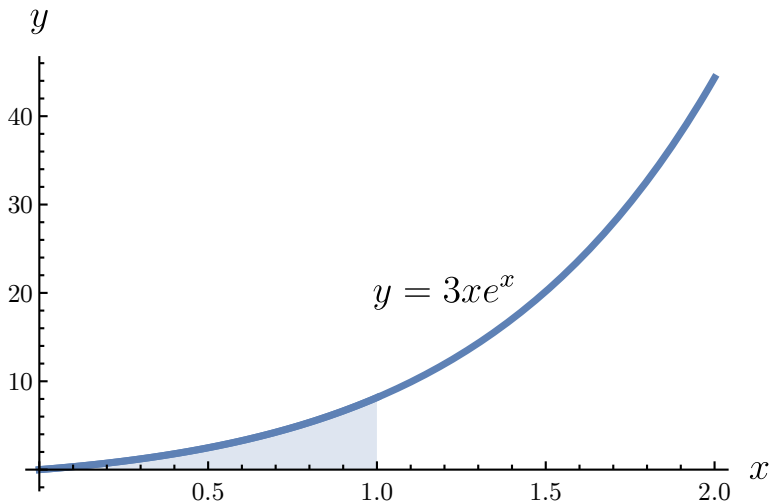
For u' and v' continuous,

$$\int_a^b u(x) \cdot v'(x) \, dx = u(x) \cdot v(x) \Big|_{x=a}^b - \int_a^b v(x) \cdot u'(x) \, dx.$$

Example #3

Find

$$\int_0^1 3xe^x \, dx.$$



$$\begin{array}{ccc}
 u's & & dv's \\
 \hline
 & & \\
 u & \xrightarrow{+} & dv \\
 du & \xrightarrow{-} & v
 \end{array}$$

maps to

$$\int u \, dv = u \cdot v - \int v \, du$$

Example #4

Find

$$\int \ln(x) \, dx.$$

Example #5

Find

$$\int x^5 \ln(x) \, dx.$$

Remark

If we have

$$\int p(x) \ln(x) \, dx$$

where p is a polynomial then choose

$$u = \ln(x) \quad \text{and} \quad dv = p(x) \, dx.$$

So differentiate the logarithm.

Example #6

Find

$$\int x^2 e^{2x} \, dx.$$

Example #7

Find

$$\int x^5 e^x \, dx.$$

Remark

If we have

$$\int p(x)e^{kx} dx$$

where p is a polynomial then choose

$$u = p(x) \quad \text{and} \quad dv = e^{kx} dx.$$

So differentiate the polynomial.

Example #8

Find

$$\int x^2 \sin(x) \, dx$$

Remark

If we have

$$\int p(x) \sin(kx) \, dx \quad \text{or} \quad \int p(x) \cos(kx) \, dx$$

where p is a polynomial then choose

$$u = p(x) \quad \text{and} \quad dv = \sin(kx) \, dx \quad \text{or} \quad dv = \cos(kx) \, dx.$$

So differentiate the polynomial.

Example #9

Find

$$\int \sin(x) e^{2x} \, dx.$$

Remark

If we have

$$\int e^{kx} \sin(mx) \, dx \quad \text{or} \quad \int e^{kx} \cos(nx) \, dx$$

apply tabular integration by parts twice, and then carefully solve for the symbol

$$\int e^{kx} \sin(mx) \, dx \quad \text{or} \quad \int e^{kx} \cos(nx) \, dx.$$

It does not matter which term you choose as u or dv , as long as you are consistently choosing the same “type” of term.

Example #10

Find

$$\int \ln^2(x) \, dx = \int \ln(x) \cdot \ln(x) \, dx.$$